Document Information

Division	Module Name
Villa Shipping	Villa Shipping | Vessels
Subject	
Vessel Monitoring System	

Version	Date	Author
.01	March 22, 2026	Juven Legaspi
Description		
This is the proposed solution design to meet the requirements for Vessel Monitoring System under the Villa Shipping Division. The purpose of this system is to replace the current manual process of recording vessel activities and voyage logs with a centralized digital monitoring system. Through this system, authorized users will be able to record, monitor, and generate reports related to vessel operations and voyage activities. The solution is intended to improve data accuracy, operational efficiency, and real-time visibility of vessel movements and activities.		

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Overview

The **Vessel Monitoring System** is designed to automate the process of tracking vessel activities and voyage logs within the **Villa Shipping Division**.

Currently, vessel operations and voyage activities are recorded manually. This manual approach can lead to:

- Inconsistent record keeping
- Delayed reporting
- Data inaccuracies
- Difficulty in tracking vessel movement history

By implementing a digital monitoring system, vessel information, voyage logs, and operational activities will be recorded and stored within a centralized system.

This will allow authorized personnel to easily monitor vessel operations, review voyage history, and generate reports when needed.

The system will also introduce structured access control to ensure that only authorized users can view or manage specific vessel records.

Objectives

The primary objectives of the Vessel Monitoring System are:

- To digitalize vessel voyage logging processes
- To improve operational transparency and monitoring
- To reduce human error caused by manual recording
- To provide a structured system for tracking vessel activities
- To allow authorized users to generate reports for operational review
- To ensure data consistency and historical tracking of vessel movements

Scope of the System

The Vessel Monitoring System will cover the following core functionalities:

1. **Vessel registration and management**
   The system will allow administrators to manage vessel information that will serve as the foundation for voyage tracking.
   - Vessel records will contain essential operational and identification details
   - Each vessel may have assigned personnel responsible for logging voyage activities
2. **Voyage Log Creation and Monitoring**
   Authorized users can create and manage voyage logs for specific vessels.
   - Each voyage log represents a single operational trip
   - Users input relevant voyage details such as cargo, location, and crew information.
3. **Activity Monitoring**
   Users can record operational activities during a voyage.
   - Activities include key operational events such as sailing, docking, anchoring, delays, and other events.
   - Activity types are configurable based on operational requirements.

- Each activity contributes to a chronological timeline of vessel operations.

4. **Status Tracking**
   The system provides predefined status indicators reflecting vessel operational conditions.
   - Status updates support real-time monitoring of vessel operations.
5. **Reporting**
   The system includes functionality to generate operational reports.
   - Reports summarize voyage logs, vessel activities, and operational performance.
   - Reports can be exported or printed for management use.
6. **Dashboard Monitoring**
   The system includes a dashboard for real-time monitoring of vessel operations.
   - Displays key indicators: total vessels, total voyage logs, active voyages, completed voyages.
   - Includes charts and summary cards for clear, easy-to-understand visualizations.
7. **System Accessibility**
   The system will be accessible through a web-based interface and integrated within the Villa Shipping Division System.

Requirements

1. **Vessel Creation and Management**
   - The system shall allow the System Administrator to create and manage vessel records.
   - Each vessel record must contain essential operational and identification details.
     - **Conditions:**
       - Only the System Administrator can create or modify vessel records.
       - Each vessel may have assigned personnel responsible for logging voyage activities.
2. **Creation of Voyage Logs**
   - Voyage logs represent the operational journey of a vessel during a specific trip.
     - **Condition**
       - Only the Captain or an authorized user assigned to the vessel can create voyage logs.
       - Each voyage log must contain operational details such as cargo information, vessel location, and crew information.
       - A unique Voyage ID must be automatically generated.
3. **Voyage Activity Management**
   - The system shall allow recording of operational activities throughout a voyage.
   - Activities represent operational events such as vessel movement, docking, sailing, anchoring, or other operational updates.
   - Each voyage log may contain multiple activity entries, creating a timeline of vessel operations throughout the voyage. Each activity entry will include:
     - Vessel operational status
     - Activity description
     - Additional remarks
     - System-recorded timestamps
     - **Condition**
       - Only the Captain or authorized users assigned to the vessel are allowed to add, update, or complete voyage activities.
       - Activities must follow the defined workflow rules for starting, ending, and completing activities.

- Once a voyage is marked as **Completed**, no additional activities may be added or modified unless authorized by the System Administrator.

4. **Activity Execution**
   - Users interact with system controls such as Add Activity, Start, End, Update, and Complete to record vessel activities.
   - This process allows the system to capture the timeline and operational status of the vessel throughout the voyage.
     - **Rules:**
       - **Adding and Starting an Activity**
         - Only the Captain or authorized user may click **Add Activity** to create a new activity record.
         - After adding, clicking **Start** will save the activity and begin tracking its duration.
         - Each activity must be linked to an existing voyage log.
         - Multiple activities may be recorded under a single voyage.
       - Ending / Completing an Activity
         - Activities must be properly marked as **Ended** or **Completed** following the system workflow.
5. **Data Reporting**
   - The system shall generate a PDF report for printing
   - The report shall include all voyage activities per voyage logged

6. **Dashboard Functionality**
   - The system shall display vessel summary information on the dashboard for management review.
   - Users can drill through from summary metrics to detailed records.
   - Dashboard data shall update dynamically based on system records.
   - Metrics displayed include total vessels, active voyages, and completed voyages.

Specifications

The Vessel shipping module in Villa system and company will include the following fields:

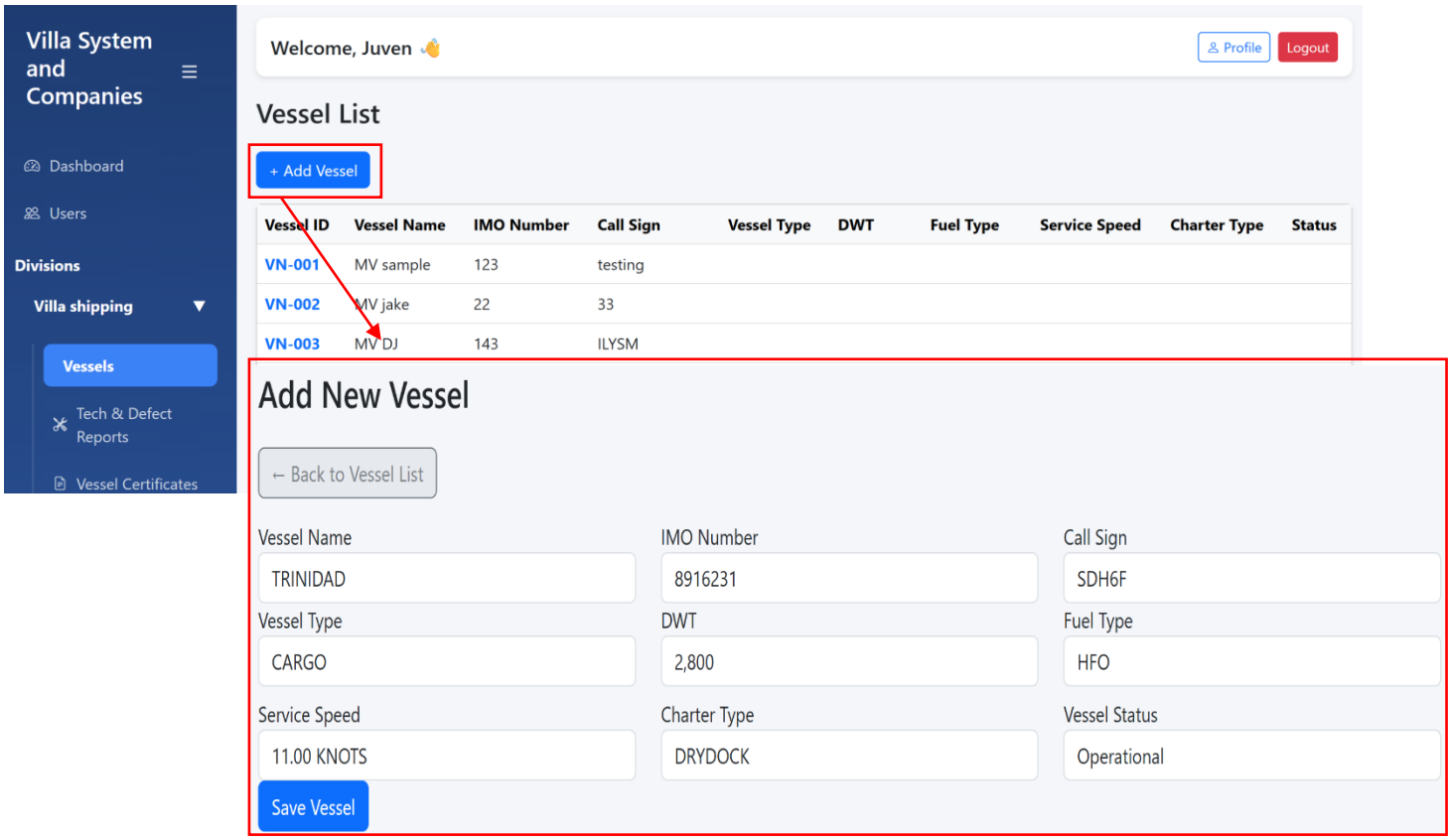
Module	Field Name	Data Type / Control	Description	Values
Vessel	Vessel ID	Auto-generated number	Unique identifier for each vessel	Cannot be modified
Vessel	Vessel Name	Text	Name of the vessel	Required
Vessel	Vessel Type	Dropdown	Type of Vessel	e.g., Cargo, Tanker, Tug, Passenger
Vessel	Service Speed	Text/Integer	Standard operating speed of the vessel	Required
Vessel	IMO Number	Text/Integer	International Maritime Organization Number	Unique, required
Vessel	DWT	Text/Integer (tons)	Deadweight tonnage of the vessel	Required
Vessel	Fuel type	Text/Integer	Type of fuel used by vessel	Required

Vessel	Charter Type	Dropdown	Type of vessel assignment	Drydock, Voyage
Vessel	Call Sign	Text	Vessel's official call sign	Optional
Vessel	Vessel Status	Dropdown	Operational condition of the vessel	Operational, Non-operational
Vessel > Voyage Logs	Voyage ID	Auto-generated number	Unique identifier for each voyage	Cannot be modified
Vessel > Voyage Logs	Cargo Volume	Text/Integer (tons)	Total cargo volume for the voyage	Required
Vessel > Voyage Logs	Cargo Type	Text	Type of cargo carried	Required
Vessel > Voyage Logs	Port Location	Dropdown	Current location of the vessel	Required
Vessel > Voyage Logs	Fuel ROB	Integer (litres)	Remaining fuel onboard at voyage start	Required
Vessel > Voyage Logs	Crew Onboard	Integer	Total number of crew members on vessel	Required
Vessel > Voyage Logs	Voyage Number	Text/Integer	Unique voyage number assigned by port	Required
Vessel > Voyage Logs	Date Created	Date (auto-populated)	Date when the voyage log is created	System-generated
Vessel > Voyage Logs > Activities	Status	Dropdown	Operational status of the vessel during activity	At Berth, Sailing, Anchored, Drydock, Delayed, etc.
Vessel > Voyage Logs > Activities	Activity Description	Text	Details of the activity performed	Required
Vessel > Voyage Logs > Activities	Remarks	Text	Additional notes or remarks on the activity	Optional
Vessel > Voyage Logs > Activities	Timestamp	System DateTime	Date and time when activity is recorded	System-generated

Expectation

1. Vessel Registration Screen

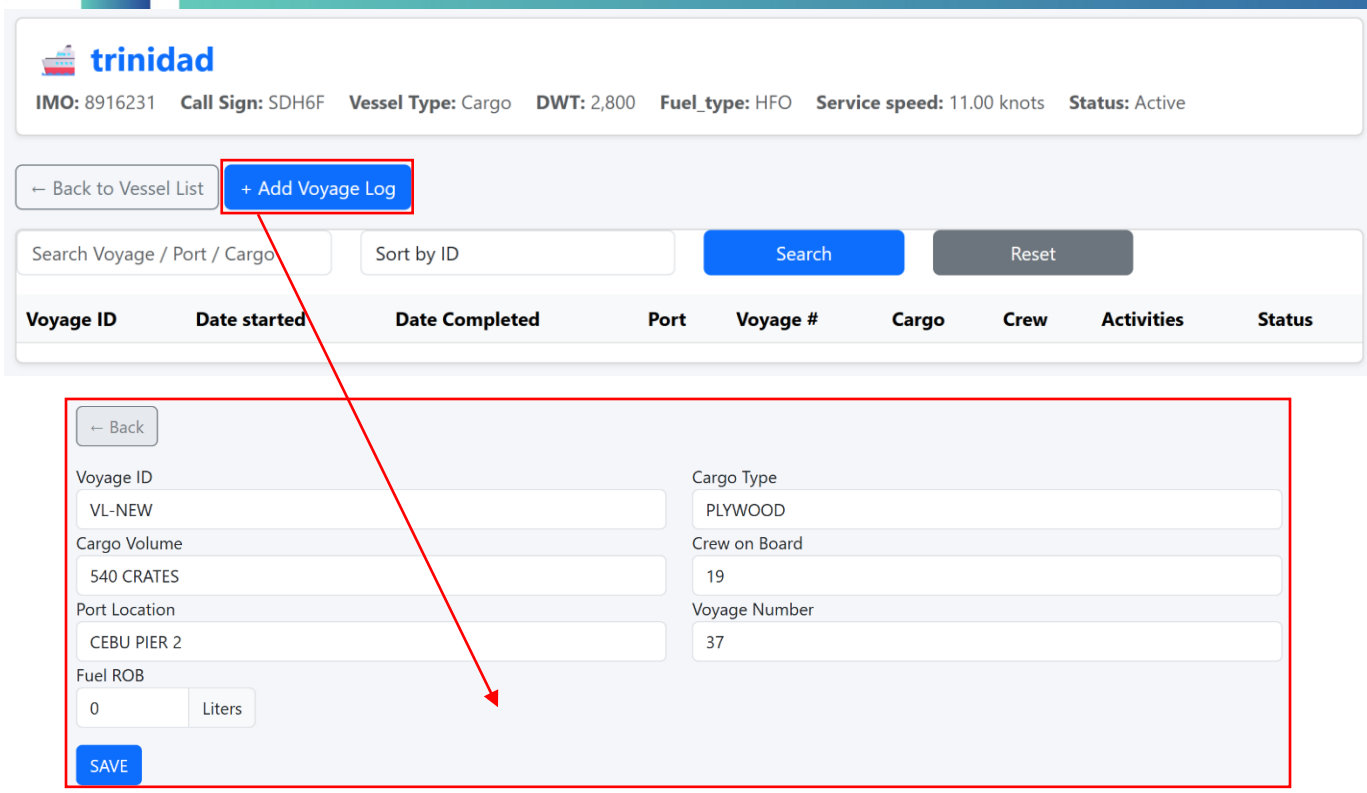
User Action	System Response	Expected Result
Click **"+ Add Vessel"**	Opens a new activity form	User can input Vessel details before clicking "Save"



The administrator can enter Vessel Name, Vessel Type, Fuel Type, and assign personnel. Upon saving, the system validates required fields.

2. Voyage Log Creation Screen

User Action	System Response	Expected Result
Click **"+ Add Voyage"**	Opens a new activity form	User can input Voyage details and can enter "Save" to save the Voyage details.



Authorized users can input cargo, port location, fuel onboard, and crew details. Voyage ID is auto-generated.

3. Activity Monitoring

User Action	System Response	Expected Result
Click "**ADD FIRST ACTIVITY**"	Opens a new activity form	User can input activity details linked to the current voyage
Click "**START**"	Activity begins tracking	Total hours start counting from this moment
Click "**END**"	Total hours are calculated	System calculates duration from start to end
Click "**END**"	End details are recorded	End time and other metadata are displayed in the activity log
Click "**END**"	"**COMPLETE**" button becomes visible	User can proceed to finalize this activity
Click "**COMPLETE**"	"**ADD**" button becomes visible	Allows user to add another activity for the same vessel
Click "**COMPLETE**"	"**COMPLETE VOYAGE**" button appears in the header	User can finalize the overall voyage for the vessel
Click "**COMPLETE VOYAGE**"	Marks voyage as completed	All activities for the voyage are finalized; no further edits allowed unless authorized

Vessels / trinidad / VL-00021

Voyage Information

Date Created	Voyage ID	Port Location	Voyage Number
2026-03-22	21	cebu pier 2	37
Fuel ROB	**Cargo Type**	**Cargo Volume**	**Crew on Board**
0liters	plywood	540 crates	19

Tracking Timeline

No activities yet.

ADD FIRST ACTIVITY

Add Activity

AT BERTH | ARRIVAL | REMARKS

START

Vessels / trinidad / VL-00021

Voyage Information

Date Created	Voyage ID	Port Location	Voyage Number
2026-03-22	21	cebu pier 2	37
Fuel ROB	**Cargo Type**	**Cargo Volume**	**Crew on Board**
0liters	plywood	540 crates	19

Users can select activity status, enter activity description, and save remarks. The system records a timestamp automatically.

Tracking Timeline

Status: AT BERTH **Activity:** ARRIVAL **Remarks:** - **Total hours:** --:--

Start: Mar 22, 2026 09:30 PM

END

DOWNLOAD PDF

Tracking Timeline

Status: AT BERTH **Activity:** ARRIVAL **Remarks:** - **Total hours:** 0.11

Start: Mar 22, 2026 09:30 PM

End: Mar 22, 2026 09:37 PM

Once "***END*** is clicked, details will be recorded and displayed.

Clicking the "***END***" button will calculate the total hours for the activity, from start to end.

UPDATE COMPLETE

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After clicking "***END***" button the "***COMPLETE***" button will become visible.

Status: AT BERTH **Activity:** ARRIVAL **Remarks:** - **Total hours:** 0.11

Start: Mar 22, 2026 09:30 PM
End: Mar 22, 2026 09:37 PM

ADD

DOWNLOAD PDF

After clicking "***COMPLETE***" button the "**ADD**" button will appear, allowing the user to add another activity for the same vessel.

Voyage Information

COMPLETE VOYAGE

Date Created	Voyage ID	Port Location	Voyage Number
2026-03-22	21	cebu pier 2	37
Fuel ROB	**Cargo Type**	**Cargo Volume**	**Crew on Board**
0liters	plywood	540 crates	19

After clicking the "***COMPLETE***" button, the "**COMPLETE VOYAGE**" button in the header will be displayed, which finalizes the overall voyage for the vessel.

Tracking Timeline

Status: AT BERTH **Activity:** ARRIVAL **Remarks:** - **Total hours:** 0.11

Start: Mar 22, 2026 09:30 PM
End: Mar 22, 2026 09:37 PM

ADD

DOWNLOAD PDF

Add Activity

AT BERTH | ARRIVAL | REMARKS

START

By clicking the "ADD" button opens a new activity pop-up form, allowing the user to create another activity for the same vessel.

✔ Voyage Completed

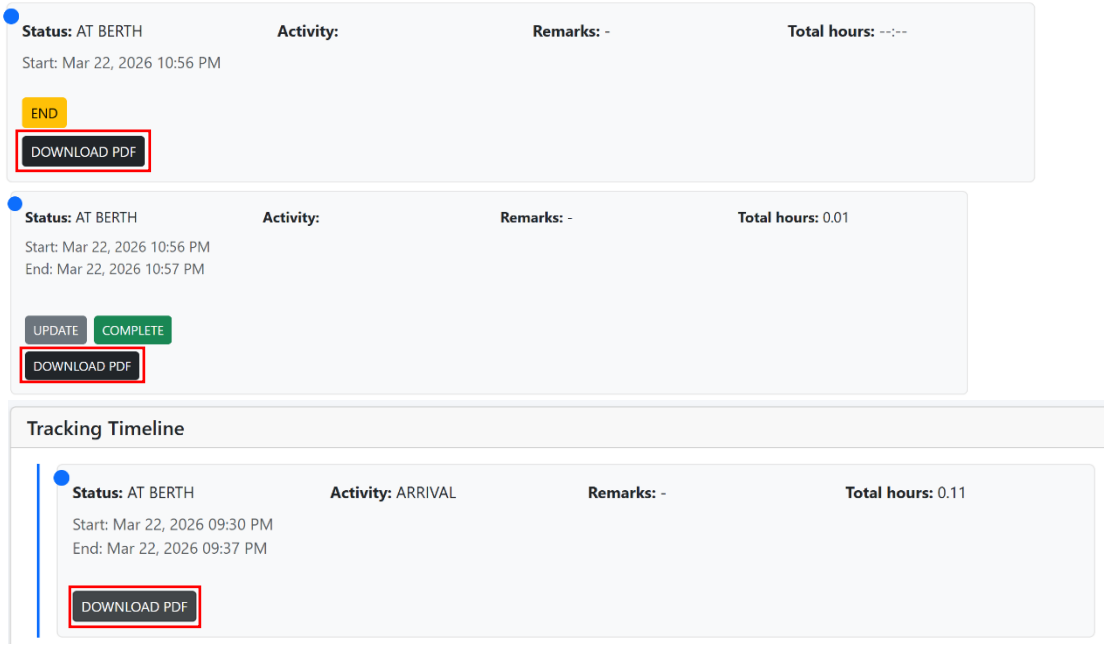
Search Voyage / Port / Cargo | Sort by ID | Search | Reset

Voyage ID	Date started	Date Completed	Port	Voyage #	Cargo	Crew	Activities	Status
VL-00021	03-22-2026	03-22-2026	cebu pier 2	37	plywood	19		COMPLETED

Clicking the "COMPLETE VOYAGE" button finalizes the voyage and all its activities. After completion, no additional activities can be added, and the voyage status will be set to "COMPLETED." To start a new voyage, a new voyage record must be created.

4. PDF Report Example

User Action	System Response	Expected Result
Click "**Download PDF**"	System generates a PDF of the selected activity or voyage	The user can download a PDF for a single activity or for the entire completed voyage. All completed activities and voyage details are included in the report



Voyage Log Report

Voyage ID: VL-00021
Voyage No: 37
Port Location: cebu pier 2
Date Started: 03-22-2026
Cargo Type: plywood
Cargo Volume: 540 crates
Crew on Board: 19
Fuel ROB: 0liters
Date Completed: 03-22-2026

Activity Timeline

Status	Activity	Start	End	Total Hours	Remarks
AT BERTH	ARRIVAL	03-22-2026 09:30 PM	03-22-2026 09:37 PM	0.11	

Prepared By:

Juven Legaspi

Date Generated:

03-22-2026

Document Approval and Acceptance

Signature **Name** **Position**	This document has been prepared by: **Juven Legaspi** System Developer	DATE
Signature **Name** **Position**	This document has been reviewed by: **Deniah Jean Placibe** System Researcher	DATE
Signature **Name** **Position**	This document has been reviewed and approved by: **Dr. Guy D. Gabison** HR Director	DATE
Signature **Name** **Position**	This document has been approved by: **Lloyd R. Bohol** Villa Shipping President	DATE